

Practical Ansys Scope for a 150 W LLC Converter Simulation Project

Which Ansys products are needed, what each one must produce, how difficult the work is, and where the scope should stop.

RECOMMENDED ANSYS STACK

Required Twin Builder/Simplorer + PExprt/Maxwell + Icepak

Conditional Q3D Extractor OR SIwave - choose one

Excluded baseline HFSS, Mechanical, live full-physics co-simulation

65-70%

project effort in Ansys

2-3 / 5

baseline difficulty

~200 h

focused engineering work

No HPC

required for baseline

01 Executive answer

The project can be strongly Ansys-centered without becoming a full multiphysics research project.

Recommended allocation: 65-70% Ansys modeling, 15% analytical design and parameter preparation, 15% hardware or reference-data correlation, and 5% reporting. Within the Ansys work, magnetics should dominate because it is the most technically distinctive part of an LLC converter.

REQUIRED

Twin Builder / Simplorer

system

PExprt + Maxwell 2D

magnetics

Icepak steady state

thermal

CONDITIONAL

Q3D Extractor

targeted loops

or SIwave

full PCB

NOT BASELINE

HFSS

radiated EMI

Mechanical

stress/life

Full live co-sim

all physics

The exact answer to “what Ansys do we need?”

Priority	Product / design type	Needed for	Decision
1	Ansys Electronics Desktop (AEDT)	Common project environment, geometry, variables, reporting and solver access	Required
2	Twin Builder or Maxwell-included Simplorer/Circuit	0D switching/system model and data handoff	Required; full Twin Builder preferred
3	PExprt + Maxwell 2D	Core/winding selection, Lm, leakage, flux and magnetic loss	Required; main technical focus
4	Icepak in AEDT	Steady-state PCB/component temperatures and heat paths	Required at compact-model level
5	Q3D Extractor or SIwave	Power-loop parasitics; choose based on available ECAD data	Conditional; only one is needed
6	Optimetrics / PyAEDT	Small parameter sweeps and automatic report export	Optional after models are stable

If only one solver is available

Choose Maxwell. Magnetic design, leakage, AC winding loss and flux distribution are the most unique Ansys contribution to an LLC project.

If two solver families are available

Choose Maxwell + Twin Builder/Simplorer. This closes the loop between converter operating waveforms and magnetics.

1 tutorial

2 guided

3 independent

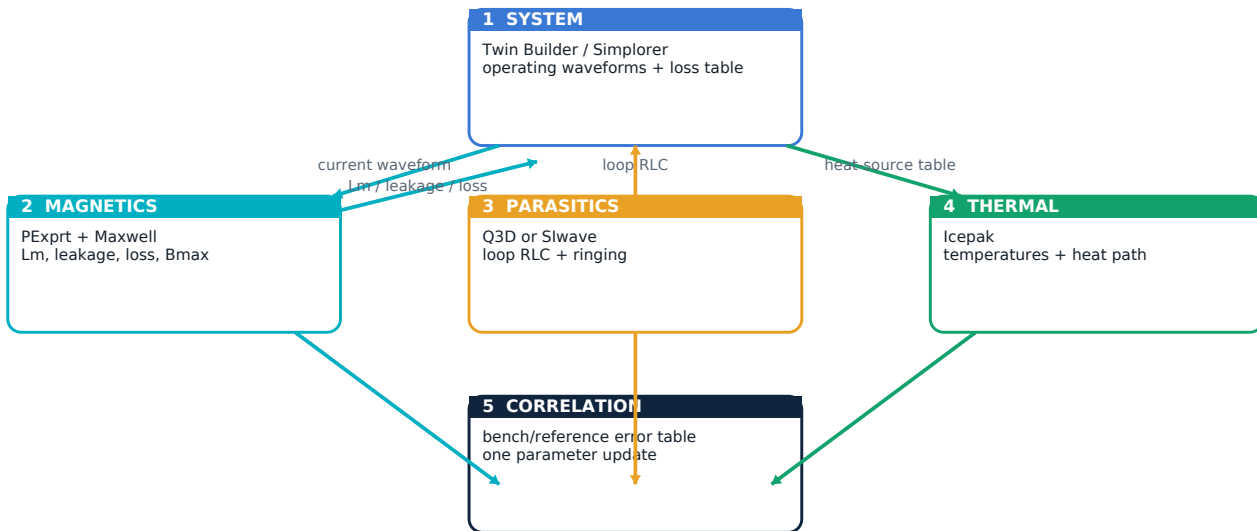
4 advanced

5 research-grade

Baseline target: difficulty 2-3. Optional work may reach 4. Difficulty 5 is explicitly excluded.

02 Recommended Ansys architecture

Use separate models with controlled data exchange. Do not start with live solver-to-solver co-simulation.



Why one-way coupling is the correct baseline

- Debuggable: every handoff is a small table of physical values rather than a hidden solver link.
- Low compute: field solvers run only when their geometry or excitation changes; the circuit model remains fast.
- Reviewable: the mentor can inspect Lm, leakage, loss and temperature assumptions independently.
- Recoverable: if one license is unavailable, the other models still produce useful deliverables.

Minimal data contract between models

From → To	Values passed	Format	Update frequency
System → Maxwell	Primary/secondary current waveform, switching frequency, duty/dead time, ambient temperature	CSV + operating-point table	At 36/48/60 V and selected loads
PExprt/Maxwell → System	Lm, leakage L, DCR, Rac(f), core loss, winding loss, equivalent capacitance estimate	Parameterized equivalent circuit	After each winding/core revision
Q3D/Siwave → System	Primary and SR commutation-loop L/R; selected capacitances; reduced SPICE/SML network	RLC table or exported model	After PCB layout freeze
System + EM → Icepak	MOSFET, magnetic, capacitor and copper heat-source values at worst case	Component power table	Nominal and worst-case scenarios
Icepak → System	Component temperatures; updated Rds(on), DCR and material-loss assumptions	One correction iteration	Once after baseline thermal solve

Scope rule

Perform one manual back-annotation iteration. A second iteration is allowed only if temperature or parasitics materially change the predicted operating point. Continuous two-way coupling is not required.

03 Twin Builder / Simplorer system model

Purpose: determine what waveforms and losses the field solvers must use - not to reproduce every transistor inside the controller IC.

Required model fidelity

Half-bridge switches; Lr-Cr-Lm tank; transformer equivalent; diode or SR rectifier; output capacitor/load; behavioral variable-frequency control; 36/48/60 V input; 10/50/100% load.

Difficulty

2.5 / 5

A guided switching model is realistic for an intern. The main difficulties are convergence, startup and defining the correct transformer/tank parameters.

Required analyses and outputs

Analysis	What to sweep	Output to save	Acceptance
Steady switching	36/48/60 V; 10/50/100% load	fs, resonant current, switch-node voltage, output voltage, RMS/peak stress	Stable periodic solution; plausible energy balance
Gain envelope	100-250 kHz; Lm/Lr and load	Gain-frequency curves and feasible operating window	Required gain reached without entering unsafe region
Soft switching proxy	Line/load corners	Current polarity and dead-time charge available before turn-on	Report margin; do not claim exact ZVS from ideal model
Startup	Soft-start frequency trajectory	Peak tank current and output overshoot	No uncontrolled current or saturation-inducing volt-seconds
Loss export	Nominal + worst case	MOSFET conduction/switching estimate, capacitor RMS, magnetic RMS currents	Heat-source table for Icepak/Maxwell

Keep the model easy

- Start with ideal switches plus explicit dead time and measured/estimated parasitics. Add one non-ideality at a time.
- Represent the UCC25600 as a behavioral frequency command and protection state machine. Do not build a transistor-level IC model.
- Use a transformer equivalent circuit first. Replace it with a Maxwell/PEprt reduced model only after the field solution is stable.
- Use fixed-step switching simulation only where necessary; use averaged/FHA curves for broad sweeps.

Estimated focused time

8 h AEDT/Twin Builder onboarding + 18 h first switching model + 12 h corner sweeps and cleanup = about 38 h. Add 20-40 h only for advanced control or live Maxwell co-simulation; these are not baseline requirements.

04 PExprt + Maxwell magnetics workflow

This should be the center of the project: roughly 40% of all Ansys effort.

Use a two-level approach. PExprt quickly searches core, material, turns, conductor and gap combinations and creates an equivalent model. Maxwell 2D then verifies the chosen geometry with higher-fidelity fields. Maxwell 3D is used only for one final winding arrangement if 2D cannot capture end effects. Ansys documents PExprt as calculating combinations of core size/material, wire gauge, turns and gap, then refining Lm, leakage, capacitance, loss and temperature rise. [5]

Stage	Solver / model	Required result	Difficulty
A. Fast sizing	PExprt analytical + embedded FEA	Core/window feasibility, turns, gap, wire, Bpeak, DCR, first loss estimate	2 / 5
B. Static verification	Maxwell 2D magnetostatic	Inductance matrix, Lm, leakage trend, flux path, saturation margin	2.5 / 5
C. AC loss	Maxwell 2D eddy-current at selected harmonics	Rac(f), proximity/skin loss, gap-fringing hotspot, winding-loss distribution	3 / 5
D. Final waveform	Maxwell transient or harmonic reconstruction	Non-sinusoidal B(t), core-loss estimate and final peak flux	3.5 / 5
E. One 3D check	Maxwell 3D eddy-current / transient	End-turn and asymmetric leakage check on final geometry only	4 / 5 - optional

Required parameter sweeps

- Air gap: enough magnetizing inductance without excessive fringing loss.
- Winding arrangement: grouped vs interleaved; compare leakage, capacitance proxy and AC copper loss.
- Conductor: litz bundle / foil equivalent; compare current density and Rac at 100, 150 and 250 kHz.
- Material and temperature: use the ferrite vendor data in the valid frequency/flux/temperature range.

Modeling traps and simple treatments

Trap	Do this	Avoid
Thousands of litz strands	Use equivalent stranded/litz representation and measured Rac correction	Meshing every strand
Core loss under square-wave excitation	Use selected harmonics or final transient waveform and vendor-calibrated loss data	Using a single 150 kHz sine result as exact truth
Leakage mismatch	Parameterize bobbin spacing and winding order; validate with LCR/impedance measurement	Treating a 2D leakage value as universally exact
Gap fringing	Apply local mesh refinement around the gap and conductor edges	Uniform coarse mesh

Estimated focused time

PExprt 10-14 h + Maxwell 2D setup/validation 35-45 h + one final refinement 10-15 h = 55-74 h. A full 3D design campaign would add 30-60 h and is not required.

05 Q3D Extractor or SIwave - choose one

The project needs parasitic extraction, but not a full EMC sign-off campaign.

Q3D extracts frequency-dependent R, L, C and G and can generate equivalent circuit models for Twin Builder/Simplorer. Ansys explicitly positions it for converter structures, bus inductance and overvoltage analysis. [6] SIwave is better when a complete ECAD layout already exists and the team wants board-level DCIR/power-integrity/EMI workflows. [7]

Question	Choose Q3D Extractor	Choose SIwave
Geometry available	A small 3D loop, busbar, package/connector or simplified copper geometry	Complete PCB database: ODB++, EDB or supported ECAD layout
Main output	Partial inductance/resistance/capacitance matrix and compact SPICE/SML model	Board DCIR, power delivery, plane current and layout-level EMI/PI indicators
Best fit here	Recommended: primary half-bridge loop + secondary SR loop	Use only if the team already has an established SIwave PCB flow
Difficulty	3 / 5 for two targeted loops	3.5 / 5 for a complete board

Minimum Q3D scope

- Extract only two loops: primary half-bridge commutation loop and secondary SR/output-capacitor loop.
- Frequency sweep: DC plus selected points spanning switching frequency to ringing range; for example 100 kHz to 30-50 MHz.
- Report loop L, AC/DC R, selected capacitance and current-density hotspots. Export a reduced equivalent circuit.
- Back-annotate the model into the system simulation and compare predicted ringing/overshoot with one measured waveform.

Do not require

- Full-board radiated EMI prediction, cable/common-mode models, chamber correlation or standards compliance.
- Both Q3D and SIwave. Duplicate extraction creates work without a clear design decision.
- Very fine package-internal bond-wire models unless the selected device package clearly dominates the loop.

Estimated focused time

4-6 h guided Q3D learning + 10-14 h geometry/terminal setup + 6-10 h matrix reduction and back-annotation = 20-30 h. A complete SIwave PCB campaign is closer to 35-60 h.

06 Icepak thermal scope

Use compact, steady-state electronics thermal modeling. The goal is design direction and correlation, not package-level CFD research.

Required model Compact 4-layer PCB; component blocks for primary MOSFETs, SR MOSFETs, transformer, resonant inductor and main capacitors; heatsinks/pads; enclosure or open-bench air; natural convection baseline.	Difficulty 3 / 5 The solver is not the hard part. The main difficulty is assigning credible heat sources, material properties and convection boundaries.
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Scenario	Boundary condition	Heat sources	Output
Baseline	25°C ambient; natural convection; open bench or defined enclosure	Nominal-load losses from system + Maxwell	Component temperatures and heat-flow paths
Worst case	High input / full load; component tolerances; reduced convection	Worst-case semiconductor, copper and magnetic loss	Thermal margin to chosen limits
One design comparison	Same boundary conditions	Same total losses	Heatsink, copper area or airflow option comparison

How to keep Icepak manageable

- Use scalar component power inputs from Twin Builder/Maxwell first. Do not begin with EM loss-map coupling.
- Model the PCB with effective in-plane and through-thickness conductivity or a compact layered model.
- Use steady state only. Add transient thermal behavior only if the mentor specifically needs warm-up time.
- Represent transformer/inductor as simple solids with separated core and winding loss regions; do not model every turn in CFD.
- Run a mesh-independence check on temperature at the main hotspot, not a global ultra-fine mesh.

Correlation target

Compare simulated and measured steady-state temperature rise at four points: primary MOSFET case, SR MOSFET case, transformer core, and winding/inductor surface. A realistic intern target is within $\pm 10^{\circ}\text{C}$ after one correction of heat-source or contact-resistance assumptions. Do not promise $\pm 2^{\circ}\text{C}$.

Estimated focused time 6 h beginner training + 14-18 h geometry/material/boundary setup + 10-14 h scenarios and correlation = 30-38 h. Live Maxwell-Icepak two-way coupling adds complexity and is not required.

07 Integration and correlation plan

The value is not “using many Ansys products.” The value is showing which prediction changed a design decision.

Required final evidence

Quantity	Predicted by	Measured / reference	Target	Design decision
Lm and leakage	PExprt + Maxwell	LCR/impedance analyzer	±10%	Gap and winding order
Resonant frequency	Twin Builder with measured L/C	Tank waveform / frequency sweep	±5%	Lr/Cr tolerance or control range
Key RMS currents	Twin Builder/Simplorer	Current probe or calibrated shunt	±10%	MOSFET, capacitor and conductor rating
Ringing frequency	Q3D/SIwave + switching model	Differential probe waveform	±15%	Loop/layout or snubber change
Component temperature rise	Icepak	Thermocouple + thermal camera	±10°C	Copper area, pad, heatsink or airflow

One-page correlation matrix format

Metric	Initial model	Measured	Error	Root cause	Updated model
Example: Lm	13.0 µH	11.8 µH	-9.2%	Gap/fringing + build tolerance	11.8 µH, winding drawing revised
Example: hotspot	78°C	87°C	+9°C	Contact resistance underestimated	Updated TIM/contact model

Good final claims

- “The model predicted the correct design trend and quantified error after physical correlation.”
- “Maxwell reduced winding alternatives from four to two before hardware construction.”
- “Q3D identified the commutation loop responsible for ringing; layout change reduced extracted L.”
- “Icepak showed that thermal interface/copper spreading mattered more than adding airflow.”

Claims to avoid

- “The simulation proves production reliability, EMI compliance or safety certification.”
- “The predicted temperature/loss is exact” without measurement uncertainty and material-model limitations.

08 Difficulty and scope boundaries

The baseline is deliberately designed around difficulty 2-3. Advanced items are optional, and research-grade items are excluded.

1 tutorial

2 guided

3 independent

4 advanced

5 research-grade

Baseline target: difficulty 2-3. Optional work may reach 4. Difficulty 5 is explicitly excluded.

Work item	Difficulty	Baseline status	Why it is difficult	Simplification
Twin Builder ideal switching model	2 / 5	Required	Basic topology and solver setup	Behavioral control; ideal switches first
PExprt preliminary magnetics	2 / 5	Required if licensed	Input specification quality	Use as screening, not final truth
Maxwell 2D core/winding model	3 / 5	Required	Material data, gap mesh, excitation choice	One core family; two winding variants
Q3D two-loop extraction	3 / 5	Conditional	Nets/terminals and matrix reduction	Two loops only
Icepak compact steady state	3 / 5	Required	Heat-source and boundary credibility	Compact solids and one comparison
Maxwell full 3D transient	4 / 5	Optional	Mesh/time-step/compute and validation	One final geometry only
PyAEDT automated DOE	4 / 5	Optional	API learning and failure handling	Automate report export after manual model works
Live Maxwell-Twin Builder-Icepak coupling	4.5 / 5	Excluded baseline	Licenses, convergence and opaque debugging	Manual one-way data handoff
HFSS radiated EMI + chamber correlation	5 / 5	Excluded	Common-mode paths, cables, environment and measurement	Only switching-loop parasitics / pre-scan

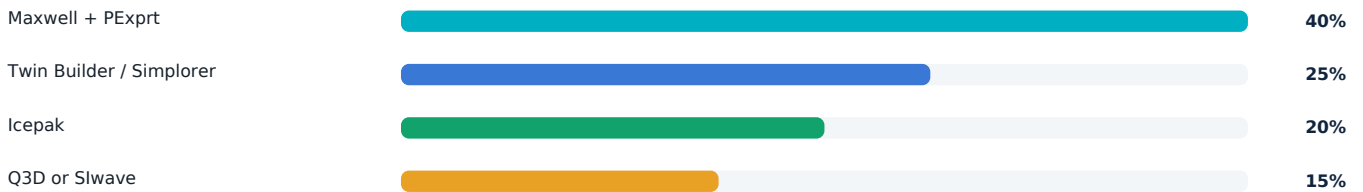
Hard stop rule

If the baseline Twin Builder, Maxwell 2D and Icepak models are not correlated by week 8, cancel all 3D, automation and co-simulation stretch goals. Finish the error analysis and reusable model templates instead.

09 Workload and learning plan

About 200 focused engineering hours produces a credible baseline; optional advanced work is separated.

Share of Ansys effort



Week	Ansys learning	Project work	Hours	Exit gate
1	AEDT + Twin Builder basics	Build topology and parameter table	18-22	Ideal switching model runs
2	PExprt + Maxwell solver selection	Core/turn/gap screening; 2D geometry	20-24	Feasible magnetic design
3	Maxwell mesh, materials and matrices	Lm, leakage, Bmax; first correlation plan	22-26	Static model converged
4	Maxwell eddy current / loss	AC copper/core-loss comparison; winding DOE	22-28	Two designs reduced to one
5	Q3D or Siwave basics	Extract two commutation loops; export RLC	18-24	Parasitics back-annotated
6	Icepak basics	Compact PCB, heat sources, natural convection	20-24	Stable steady-state model
7	Icepak refinement + reporting	Worst case + one cooling comparison	18-22	Thermal margins documented
8	Parameter export and model update	Bench/reference correlation and one back-annotation	20-26	Error matrix complete
9	Optimetrics/PyAEDT only if stable	Small sweep or final 3D check	16-24	One optional improvement
10	Reusable templates and presentation	Clean models, instructions, final review and demo	18-22	Another engineer can rerun it

Recommended official learning sequence

- Twin Builder Platform: Basics - run library models, plots and design exploration. [14]
- Maxwell Getting Started - 2D/3D setup, solver selection, mesh, parameters and reports. [15]
- Q3D Getting Started - terminals/nets, RLGC extraction and circuit back-annotation. [16]
- Icepak AEDT Getting Started - geometry, boundary conditions, mesh, solve and post-processing. [17]
- Free Maxwell-Icepak electrothermal course only after both standalone models work. [18]

Expected level after 10 weeks

Independent beginner-to-intermediate in four bounded workflows, not an Ansys multiphysics expert. That is a credible and defensible internship outcome.

10 License, installation and computing requirements

Confirm actual entitlements before week 1; the product names in the GUI do not guarantee solver licenses are available.

License checklist for the team / IT

Check	Minimum acceptable answer	Why
AEDT version	Team-standard supported release; 2026 R1 is not mandatory	Avoid incompatible project files and material libraries
Maxwell	2D + 3D solve entitlement	2D is baseline; 3D is optional final check
Circuit / Simplorer / Twin Builder	At least the Maxwell-included Simplorer subset; full Twin Builder preferred	Ansys states Maxwell includes Simplorer with a subset of Twin Builder capability. [1]
PExprt	Confirm launch and save access	Reduces magnetic sizing effort; not assumed from GUI alone
Icepak	AEDT Icepak steady-state solve	Thermal license may be separate from Maxwell/Q3D entitlement
Q3D or Siwave	One usable seat during weeks 5-6	Only one parasitic tool is required
HPC	No extra HPC required for baseline	Standard products include limited parallel cores; schedule optional 3D runs accordingly. [9]

Recommended workstation for this bounded scope

Resource	Comfortable local target	Notes
CPU	8-16 modern CPU cores	Solver use is limited by license; strong per-core performance still helps
RAM	32 GB minimum; 64 GB preferred	64 GB is useful for final 3D/Q3D/Icepak cases
Storage	200 GB free local SSD	Ansys electromagnetic installation and result files are large; official 2025 R2 electromagnetics install is ~51 GB. [12]
GPU	Recent workstation GPU; no GPU-compute requirement	Useful for 3D visualization; baseline solvers are CPU/RAM dominated
Shared compute	Optional for one 3D or refined Icepak case	Not a dependency for completing the project

Important limitation

The Ansys Electronics Desktop Student version is useful for tutorials but not a safe basis for the final project: current published restrictions include 2,000 triangles for 2D, 64,000 volume elements for 3D, 512k Icepak elements, 50 Circuit components and 15 Twin Builder components. Use the company license for the deliverable models. [10]

Mentor decision needed before starting

Confirm: (1) which system simulator is preferred, (2) whether PExprt is licensed, (3) Q3D versus Siwave based on the team workflow, (4) access to Icepak, and (5) a weekly reviewer for magnetic assumptions. Nothing else blocks the baseline.

11 Recommended final project statement

A concise scope that the Ansys team can approve without committing to an unbounded multiphysics program.

Proposed title
Ansys-Based Multiphysics Design and Hardware Correlation of a 150 W Low-Voltage LLC Resonant Converter

Project objective

Develop a reusable, parameterized Ansys workflow that predicts the system operating envelope, magnetic component behavior, critical PCB parasitics and steady-state temperature of a 36-60 VDC to 12 V / 150 W LLC converter. Validate selected predictions using low-voltage hardware or approved reference measurements, then document simulation error and one design improvement.

Must deliver 1. Twin Builder/Simplorer operating-point model 2. PExprt + Maxwell 2D magnetic workflow 3. Q3D or SIwave two-loop extraction 4. Icepak steady-state thermal model 5. One manual back-annotation iteration 6. Correlation/error matrix 7. Reusable templates and instructions	Will not promise 1. Mains PFC hardware 2. Production EMI/safety compliance 3. Full live multiphysics co-simulation 4. HFSS radiated-EMI prediction 5. Detailed thermo-mechanical fatigue life 6. Research-grade 3D litz-strand modeling 7. Digital-controller IC transistor modeling
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Success criteria

- All required models rerun from documented inputs without manual repair.
- Magnetic parameters within $\pm 10\%$, resonant frequency within $\pm 5\%$, key RMS currents within $\pm 10\%$, ringing frequency within $\pm 15\%$, and thermal rise within $\pm 10^{\circ}\text{C}$ - or the gap is technically explained.
- At least one design decision changes because of the Ansys results: winding order, gap, copper loop, heatsink/pad, copper spreading or control frequency range.
- The team receives a short “when to use which Ansys solver” guide and reusable report templates.

One-sentence pitch to the Ansys team
“I propose a low-risk Ansys workflow project in which Twin Builder defines operating waveforms, PExprt/Maxwell designs the magnetics, Q3D or SIwave captures only the critical PCB loops, and Icepak verifies thermal margin, with each model independently reviewable and correlated once against low-voltage measurements.”

12 Sources and Ansys documentation

Accessed 23 August 2026. Official Ansys/Synopsys sources are used for product capabilities and current release context.

- 1 Ansys Maxwell product page - 2026 R1, solver/circuit integration and Simplorer subset.
<https://www.ansys.com/zh-cn/home/products/electronics/ansys-maxwell>
- 2 Maxwell to Twin Builder coupling documentation.
<https://ansyshelp.ansys.com/public/Views/Secured/Electronics/v252/en/Subsystems/Maxwell/Content/Maxwell/CouplingMaxwelltoTwinBuilderComponent.htm>
- 3 Twin Builder coupling components supported across Maxwell, Q3D, Icepak and other products.
<https://ansyshelp.ansys.com/public/Views/Secured/Electronics/v252/en/Subsystems/TwinBuilder/Content/TwinBuilderCoupling.htm>
- 4 Twin Builder product page - system models, ROMs and co-simulation context.
<https://www.ansys.com/en-gb/products/digital-twin/ansys-twin-builder>
- 5 PExprt static coupling - core/material/wire/turn/gap search and magnetic/thermal equivalent models.
<https://ansyshelp.ansys.com/public/Views/Secured/Electronics/v252/en/Subsystems/TwinBuilder/Content/PExprtStaticCouplingComponents.htm>
- 6 Ansys Q3D Extractor - frequency-dependent RLCG extraction and converter equivalent models.
<https://www.ansys.com/zh-tw/products/electronics/ansys-q3d-extractor>
- 7 Ansys SIwave - complete PCB power integrity, DCIR, EMI and electrothermal workflow.
<https://www.ansys.com/Products/Electronics/ANSYS-SIwave>
- 8 Ansys Icepak - electronics/power-electronics CFD thermal modeling and 2026 R1 workflow.
<https://www.ansys.com/en-gb/home/products/electronics/ansys-icepak>
- 9 Ansys HPC documentation - included parallel cores by solver family.
https://ansyshelp.ansys.com/public/Views/Secured/corp/v242/en/licensing/HPC_detail.html
- 10 Ansys Electronics Desktop Student restrictions.
<https://www.ansys.com/academic/students/ansys-electronics-desktop-student>
- 11 Ansys Electronics Desktop platform - Maxwell, Q3D, SIwave, Icepak integration.
<https://www.ansys.com/products/electronics>
- 12 Ansys installation disk and memory requirements (2025 R2 documentation).
https://ansyshelp.ansys.com/public/Views/Secured/corp/v252/en/installation/unix_dskspacereqt.html
- 13 Ansys 2026 R1 release highlights.
<https://www.ansys.com/ja-JP/products/release-highlights>
- 14 Getting Started with Ansys Twin Builder Platform: Basics.
<https://innovationspace.ansys.com/certifications/courses/getting-started-with-ansys-twin-builder-platform-basics/>
- 15 Ansys Maxwell Getting Started.
<https://innovationspace.ansys.com/product/ansys-maxwell-getting-started/>
- 16 Getting Started with Ansys Q3D Extractor for EMC.
<https://innovationspace.ansys.com/product/getting-started-with-ansys-q3d-extractor-for-electromagnetic-compatibility/>
- 17 Electronics Thermal Management using Icepak in AEDT: Getting Started.
<https://innovationspace.ansys.com/product/electronics-thermal-management-using-ansys-icepak-software-in-ansys-electronics-desktop-aedt-platform-getting-started/>
- 18 Free electrothermal course using Maxwell and Icepak.
<https://innovationspace.ansys.com/product/etm-using-ansys-maxwell-and-icepak/>

Engineering note: software entitlements, license pools and installed versions vary by organization. The report therefore specifies solver capabilities and an entitlement checklist rather than assuming a particular commercial bundle name.